

# DONOVAN EYER

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## SKILLS

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**Skills: Python, Java, C++, OCaml, Go Programming, Flask, Git, GitHub, Windows, Linux, Ubuntu, Unix, SQL, HTML, CSS, JavaScript**

Excellent written and verbal communication, with a focus on clear and concise prose. Strong collaboration and teamwork abilities, shown through group projects and extracurricular activities. Decoding and debugging malfunctioning code.

## EXPERIENCE

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### Writing Assembly Code

- **Writing Interpreted and Compiled Code**
- Wrote code that could perform a wide range of functions, such as randomized grouping and basic artificial intelligence
- Leveraged strong programming skills to develop robust and intuitive solutions that addressed complex problems
- Actively collaborated with groups to solve complex problems and design complicated, functional applications
- Developed a comprehensive understanding of software development principles and compiled code
- Adapted differing coding methodologies to various project requirements
- Displayed flexibility in optimizing code for performance, readability, and maintainability
- Developed thorough understanding of the underlying rules and functions of assembly language
- Learned how to effectively translate assembly code into its C++ equivalent
- Identifying and properly utilizing x64 registrars in complex algorithms, resulting in efficient performance and streamlined pipelines
- Took the lead in designing efficient loops to iterate through code, optimized to minimize overhead and maximize kernel efficiency
- Resolved challenging issues by troubleshooting code, identifying and resolving bugs and malfunctioning functions

### Azure Database Designer

- In charge of developing virtual pipelines that translated SQL data into readable spreadsheets
- Designed and implemented efficient data extraction and transformation processes that converted complex databases into user-friendly formats
- Demonstrated an eye for visual design by creating virtual layouts that conveyed complex information in visually eligible ways, facilitating decision-making processes
- Communicated with teammates to analyze growth rate of numerous policies
- Utilized statistical analysis and data visualization to identify consistent trends and patterns
- Learned how to share and modify digital information across several different sub-projects
- Facilitated seamless information exchange and maintain consistent updates across disparate workstreams
- Adapted to working with diverse tools and technologies to optimize project coordination

(Python, OpenGL)

### Computer Graphics

- Wrote algorithms that create two and three-dimensional shapes using methods such as line rasterization, scan fill and
- Stored object vertices through methods such as vertex buffer object and element buffer object

- Implemented shaders and light sources to illuminate and cast shadows on 3D objects
- Used problem-solving to design code that allowed objects to automatically animate and perform cyclical functions

#### Compiled High-Language Programming

- Wrote code in Golang using Go routines to separate running threads, and channels to store the results of the separate threads in variables
- Learned about the different types of consistency (Causal, Eventual, etc.) that programs could achieve to improve aspects such as latency and availability
- Sent tasks between different servers using Remote Procedure Calls (RPCs)
- Implemented a variety of different clocks, such as Lamport and vector, in my code to improve the latency of my functions
- Designed a distributed consensus Raft program that was able to send messages and tasks between different servers, and that could be implemented with other programs

### EDUCATION

#### Boston University

Boston, Massachusetts

Bachelor's Degree, Computer Science

Sep 2021 - May 2025

- Relevant Coursework: Intro to Computer Science 1 & 2, Computational Systems, Intro to Algorithms, Combinatorial Structures, Data Mechanics, Geometric Algorithms, Intro to Computer Graphics, Concepts of Programming Languages
- GPA: 3.46

### PROJECTS

Data Pipeline Designer, Nov 2022 - Nov 2022

#### Group Project, Hackathon (BostonHacks), Boston, Massachusetts

- Collaborated with a student team to design and develop a program that scraped YouTube data and used it to categorize videos
- Extracted and analyzed data, implement machine learning algorithms, and create a user-friendly interface
- Demonstrated efficient teamwork within a fast-paced, competitive environment
- In charge of designing the pipeline code that rewrote the Python code to function on the group website

Programmer, Jul 2022 - Jul 2022

#### Personal Project, Group Randomizer, Nampa, Idaho

- Modeled and programmed a Python program that takes a list of names and randomly sorts them into groups
- Wrote data structures and algorithms to ensure fair and unbiased group assignments
- Created a solution that ran in an efficient runtime

Front-End Designer, Feb 2024 – May 2024

#### Group Project, RoomieConnect, Boston, Massachusetts

- Designed the HTML interface for the user's information page
- Wrote JavaScript code for UI elements, including buttons and pop-up windows
- Implemented SQL databases that used the host's personal information (email, personal preferences, etc.) to find other users with similar preferences